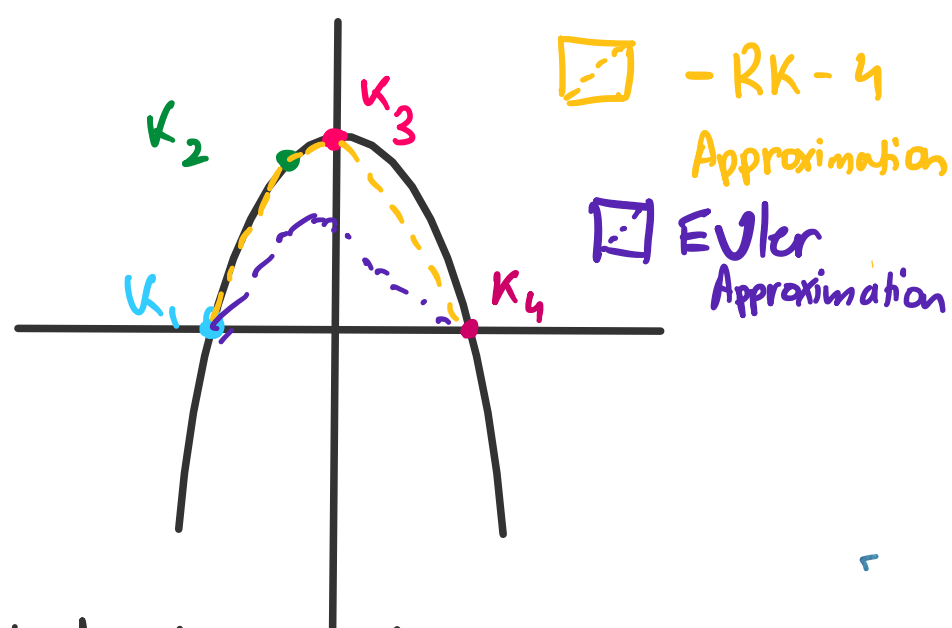


RK-4 (NOTES)

Monday, 20 July 2026

11:11 PM

Example graph:



-Instead of computing acceleration only once, RK-4 computes it at 4 strategically chosen points, K_1, K_2, K_3 , and K_4 , making it much more accurate than Euler.

Specifically, given the differential equation $\frac{dy}{dx} = f(x, y)$, Euler's method would yield:

$y_{\text{new}} = y + f(x, y) \cdot h$, where h is the step size

RK-4 however, samples the slopes 4 times, at:

$$K_1 = f(x, y),$$

$$K_2 = f\left(x + \frac{h}{2}, y + \left(\frac{h}{2}\right)K_1\right)$$

$$K_3 = f\left(x + \frac{h}{2}, y + \left(\frac{h}{2}\right)K_2\right)$$

$$K_4 = f(x+h, y + h \cdot K_3)$$

K_1 is the slope at the start

K_2 is the slope at the ESTIMATED midpoint using K_1

K_3 is the slope at the ESTIMATED midpoint using K_2

K_4 is the slope at the ESTIMATED endpoint using K_3

Then, a weighted average is taken:

$$y_{\text{new}} = y + \left(\frac{h}{6}\right)(K_1 + 2K_2 + 2K_3 + K_4)$$

$$x_{\text{new}} = x + h$$

The weighting is (1, 2, 2, 1) and $1/6$

So that RK-4 matches a certain Taylor series expansion.

Since RK-4 takes more calculations at different points on the graph and takes an average, it is far more accurate than Euler's method.